

A Structure-Agnostic SHM Decision Framework for Reusable CFRP Spacecraft Structures

from Real-Data Detection to Physics-Grounded Life Prognosis

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Keio University / Composites Part B (draft)

The reusable-vehicle decision

Reusable launch vehicles change the SHM question:

- not “did it survive one flight?”
- but **may it fly again — and for how many flights?**

CFRP primary structures (interstage, fairing, motor case, tank, drive shaft) fail by *internal* delamination / debond — invisible, inferred from NDT.

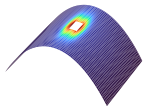
Gap: no end-to-end pipeline that propagates uncertainty from detection to a re-flight decision *and* states what real data supports.

detect → classify → characterise
→ prognose → clear
→ fleet-learn → design

One decision core, five structures, four load modes

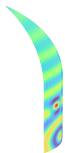
A structure-agnostic SHM decision framework across five reusable CFRP spacecraft structures and four load modes

Perforated interstage



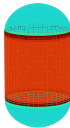
load: in-plane / membrane
sensing: surface-stress GNN

H3 fairing



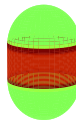
load: transverse press. / bending
sensing: guided-wave GNN

SRB-3 motor case



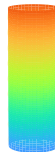
load: internal-pressure burst
sensing: acoustic emission

CFRP COPV tank



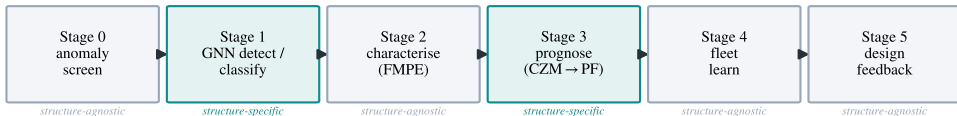
load: internal pressure
sensing: shared |z| screen

CFRP drive shaft



load: TORSION (shear)
sensing: shared |z| screen

Shared structure-agnostic decision core --- only detection (Stage 1) & prognosis physics (Stage 3) are structure-specific (pluggable)



detect → classify → characterise → prognosis → clear (OK/REPAIR/RETIRE) → fleet-learn → design | 1,499 real-FEM states, one margin surrogate $R^2=0.94$

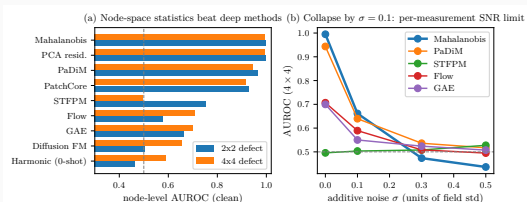
- Shared, structure-agnostic decision core (Stage 0/2/4/5 + clearance).
- Only *detection* (Stage 1) and *prognosis physics* (Stage 3) are structure-specific — pluggable.

The five structures span the load-mode space

structure	load mode	detection front-end	evidence
interstage	membrane	surface-stress GNN	FEM + meas. TSA
fairing	transv./bending	guided-wave GNN	FEM + real OGW
SRB-3 case	pressure burst	acoustic emission	FEM + real AE
drive shaft	torsion	shared $ z $ (future)	CZM-FEM
COPV tank	int. pressure	shared $ z $ (future)	FEM

- Three demonstrated end-to-end with **real data**; two extend the load-mode coverage with FEM prognosis.

Detection front-ends — grounded in real data



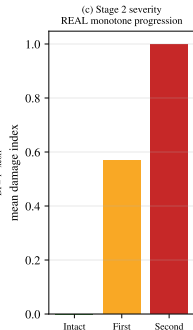
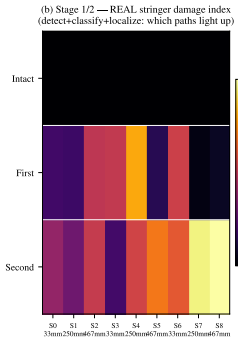
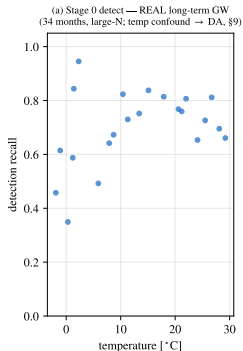
Stage-0 $|z|$ screen: per-node statistic dominates (AUROC ≈ 0.99 , fixed geometry).

Structure-specific Stage-1 detectors:

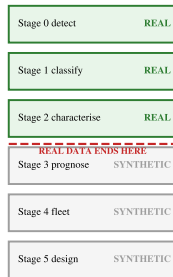
- interstage: surface-stress GNN, macro-F1 0.76 (vs GAT 0.61); detection generalises, localisation collapses OOD.
- fairing: guided-wave GNN, F1 0.86.
- SRB-3: real AE, per-hit AUROC 0.89 / per-specimen 1.00.

How far does real data carry the pipeline? (honest)

Fairing (CFRP stringer) end-to-end on REAL Open-Guided-Waves data: detect → classify → characterise are real; prognosis is honestly synthetic

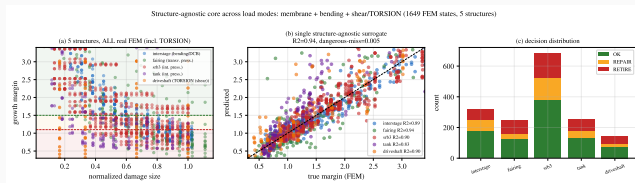


(d) honest real-data coverage



- Detect / classify / characterise on the fairing are **real data** (long-term guided waves, impact wavefield).
- **synthetic** from Stage-3 prognosis onward (representative phase-field, synthetic fleet) — the life side is the largest open gap.

Structure-agnostic core — and its boundary

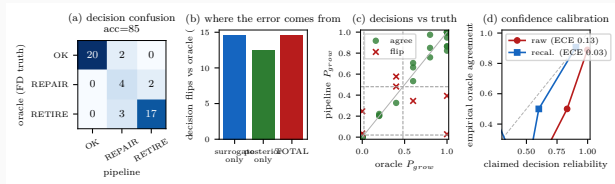


A single margin surrogate over **1,499 real-FEM** critical-load states fits all five ($R^2=0.94$, dangerous-miss 1.1%).

Honest boundary: leave-one-load-mode-out to the unseen torsion shaft *fails* ($R^2 < 0$).

Geometry transfers; load mode does not.

End-to-end decision reliability (vs a **synthetic** oracle)

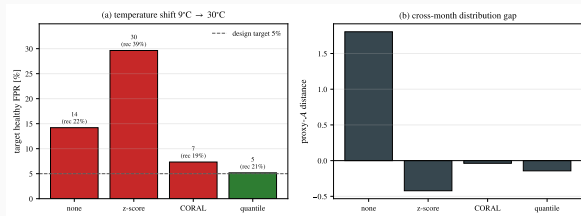


Propagate uncertainty to the clearance call; score against the exact FD-physics oracle.

- accuracy **85%**, dangerous-miss **0%**
- Platt recalibration: ECE 0.18 $\rightarrow$ 0.06

Measured against a **synthetic oracle** — internal error propagation, not agreement with real life tests.

Sim-to-real: normalisation choice controls false alarms

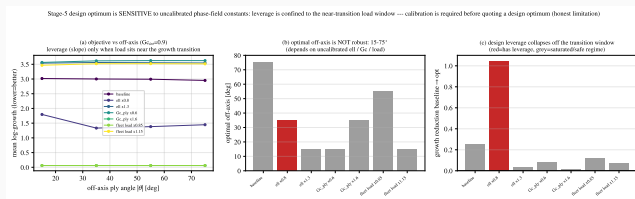


Real OGW temperature shift 9° → 30°C:

- uncorrected FPR **14%** (temperature mimics damage)
- quantile DA → **5%** (design target)
- **z-score normalisation worsens to 30%**

Interstage TSA: quantile DA closes 97% of the gap.

Stage-5 design feedback — mechanism, not a calibrated value



Fleet-driven inverse design steers off-axis angle + interface toughness.

Honest limitation: under uncalibrated phase-field constants the optimal angle spans 15°–75° — leverage exists only near the growth transition.

The **mechanism** is robust; a quoted optimum needs calibration.

- **real data**: detection / classification / characterisation (guided waves, AE, TSA), posterior coverage, 1,499-state FEM core.
- **synthetic**: prognosis, clearance, fleet, design — representative phase-field + synthetic fleets; decision UQ vs a synthetic oracle.
- **Largest gap**: no real life-test validation of the life side.
- Priority inputs: Kojima TSA full-field (interstage), real fatigue data.

- The differentiable, fast forward (FNO; autograd-exact) is what a TMCMC / Bayesian-inverse phase needs.
 - Its physics — anisotropic phase-field seeded by interlaminar cohesive zones — is the prognosis layer the torsion shaft extends from mode-I to mode-II/III shear.
- ⇒ **Phase-field** × **GNN prognosis**, calibrated on measured data, is the computational-solid-mechanics continuation.

One decision core across four load modes.

Real where data permits, honest where it does not.

Closing the life-side real-data gap is next.

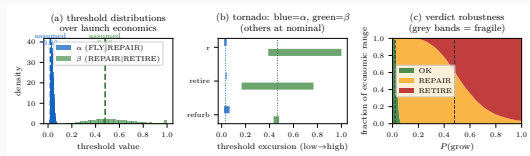
Appendix / backup

A1 — Stage-0 anomaly benchmark (clean, fixed geometry)

method	2×2	4×4
per-node Mahalanobis	0.999	0.995
PCA residual	0.999	0.995
PaDiM	0.964	0.944
PatchCore	0.927	0.918
STFPM	0.754	0.496
graph autoencoder	0.663	0.700
flow (RealNVP)	0.578	0.707
diffusion (FM)	0.503	0.663
harmonicity (0-shot)	0.465	0.590

node-level AUROC. Per-node statistics dominate on fixed geometry; the defect sits below 0.1σ (a per-measurement SNR limit).

A2 — Decision rule and launch economics



$a^* = \arg \min_a \mathbb{E}_{p(\theta|x)} [C(a, \theta)]$, actions {OK, REPAIR, RETIRE}.

- MC over refurb/loss 1/100–1/30, retire/loss 0.1–0.4
- $\alpha = 0.035$ [0.021, 0.055], $\beta = 0.53$ [0.28, 0.93]
- REPAIR-vs-RETIRE flips at $P_{\text{grow}} \approx 0.3$ –0.5 (economics-fragile)

A3 — Stage-2 characterisation (FMPE) vs differentiable physics

set	method	c_x RMSE	size RMSE	90% cov.
in-dist	FMPE	0.006	0.002	0.958
in-dist	diff-phys	0.001	0.003	1.000
OOD	FMPE	0.144	0.096	0.000
OOD	diff-phys	0.002	0.001	0.958

Forward physics generalises; amortised inference extrapolates poorly $\Rightarrow$ hybrid Stage-2 (amortised in-dist, physics-grounded OOD).

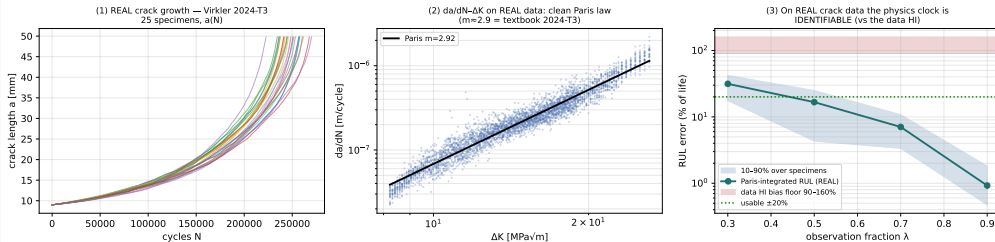
A4 — Stage-3 prognosis: anisotropic phase-field

$$\Psi(\mathbf{u}, d) = \int_{\Omega} \left[g(d) \psi^+(\boldsymbol{\varepsilon}) + \frac{G_c}{c_w} \left(\frac{d^2}{\ell} + \ell \nabla d \cdot \mathbf{A} \nabla d \right) \right] d\Omega$$

- $\mathbf{A} = \mathbf{I} + \beta \mathbf{a} \otimes \mathbf{a}$: fibre-aligned crack resistance; interface $G_c^{\text{int}}/G_c^{\text{ply}} \approx 0.3\text{--}0.6$ (DCB/ENF anchored).
- Carrara-2020 fatigue history degrades G_c across flights (no i.i.d. shortcut).
- Pluggable: fairing skin-core debond ($G = \kappa \sigma^2 a^2$, Paris); SRB-3 burst; *torsion* interlaminar CZM.

A4b — Does a remaining-life clock exist? (real-data validated)

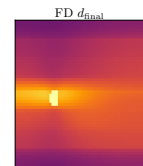
Physics life-clock validated on REAL crack-growth data (Virkler 2024-T3, 25 specimens) — identifiable RUL where the data-driven HI is not



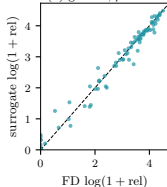
- **Data-driven** health index: *no* usable clock — point-RUL bias floor 90–160% of life (real ReMAP DFOS); no reframing rescues it.
- **Physical** Paris/phase-field clock: identifiable, validated on *real* crack growth — Virkler 2024-T3 ($m=2.92$), RUL error 16.7% $\rightarrow$ 0.9% as λ grows.
- Real CFRP DCB: identifiable per-specimen but coarser — fibre bridging breaks the single da/dN – ΔG law ($m \approx 17$).
- Same estimator: fails on the data index, works on the physics clock $\Rightarrow$ the life-side limit is the **signal**, not the method.

A5 — Fast surrogate + trust gate

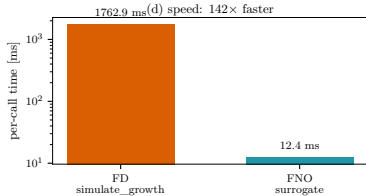
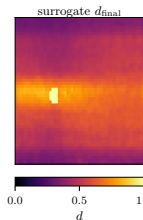
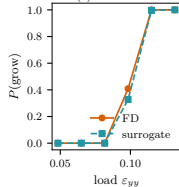
(a) load = 0.140



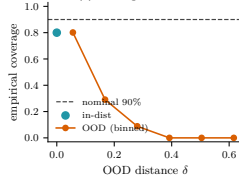
(b) growth, $\rho = 1.00$



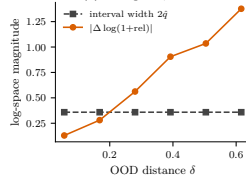
(c) calibration



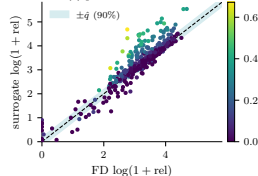
(a) coverage breaks OOD



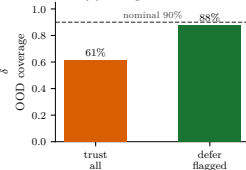
(b) error grows, width flat



(c) prediction $\pm$ interval



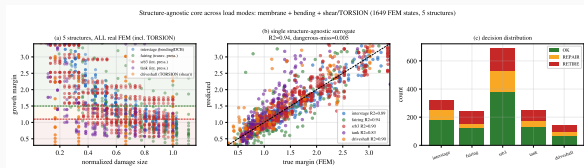
(d) trust gate: 33% $\rightarrow$ FD



Split-conformal trust gate: flags 33% of an OOD stream,
restores coverage 61% $\rightarrow$ 88%.

FNO: field rel- L_2 3.5%, grow-flag 0.988, **142** $\times$ speed.

A6 — Structure-agnostic surrogate: details



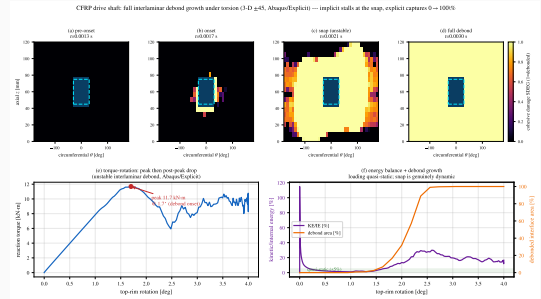
1,499 real-FEM states; descriptors [dmg, crit, aux, stiffness] + structure one-hot.

- per-structure R^2 : interstage 0.91, fairing 0.94, SRB-3 0.93, tank 0.71, drive shaft 0.91
- shared rule: acc 0.86, dangerous-miss 1.1%
- LOSO torsion: $R^2 = -1.4$, dangerous-miss 17% (load-mode boundary)

A7 — Torsion shaft: CGAX screen + 3-D explicit CZM

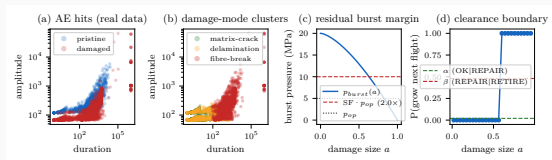
CGAX screen (± 45 tube, twist DOF):

- analytic check: wall shear 17.1 vs $T/2\pi R^2 t = 16.8$ MPa
- 144-state groove sweep, T_{cr} 0.31 – 7.76 kN·m
- true torsional buckling is non-axisymmetric $\Rightarrow$ 3-D



3-D explicit: KE/IE 1.3–3.6% in loading (quasi-static), peak 11.7 kN·m @ 1.7° , debond $0 \rightarrow 100\%$.

A8 — SRB-3 acoustic-emission detection (real data)

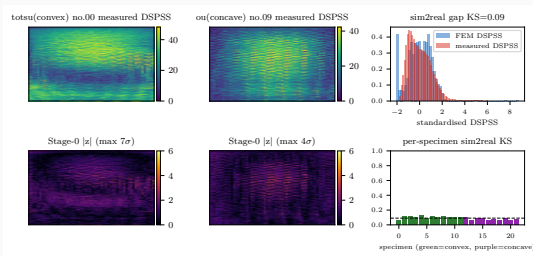


4TU CC0 AE, ~48k Vallen hits:

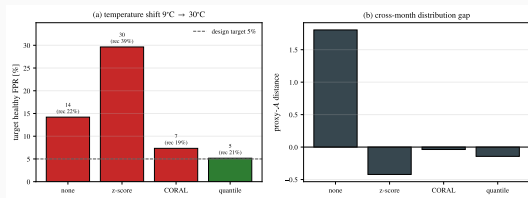
- group-aware RF: per-hit AUROC 0.888 / per-specimen 1.000 (linear baseline ~0.52)
- amplitude–duration damage-mode clustering

Honest: compression-pristine coupons also emit high-energy AE $\Rightarrow$ the multivariate detector, not one cluster, is the read-out; AE is a real-data *proxy* for the motor-case front-end.

A9 — Sim-to-real domain adaptation: details

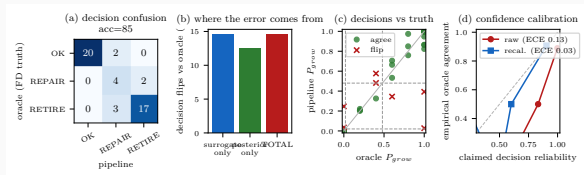


Interstage TSA: proxy- $\mathcal{A}$ 2.0 $\rightarrow$ 0.3 (97% closed).



Fairing OGW: FPR 14 $\rightarrow$ 5% (quantile), CORAL 7%, z-score $\rightarrow$ 30%; threshold set on source healthy (no target leak).

A10 — End-to-end decision UQ: error decomposition

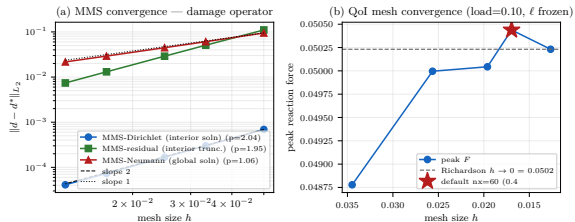


Four decisions isolate each error (over 40 held-out scenarios):

- Stage-2 posterior: flip rate 25% (dominant)
- surrogate forward: 17.5%
- partial cancellation → total 15%
- all 16 RETIRE cases caught (0% dangerous-miss)

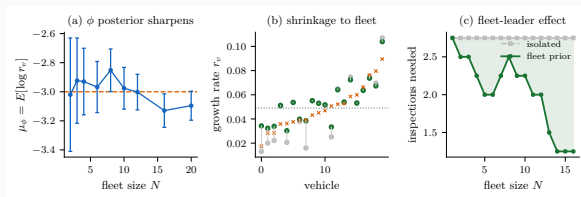
Oracle = exact FD physics (**synthetic**).

A11 — Solver verification and robustness



- MMS: interior order 2.04 (2-D), 2.05 (3-D)
- mesh convergence: 0.4% at $n_x=60$
- constant-peak proxy under-predicts life $3.4\times$ vs a variable flight spectrum (S–N anchored)
- Stage-0 holds AUROC 0.84 at $\sigma=0.1$; chance near $\sigma \approx 0.7\text{--}1.0$

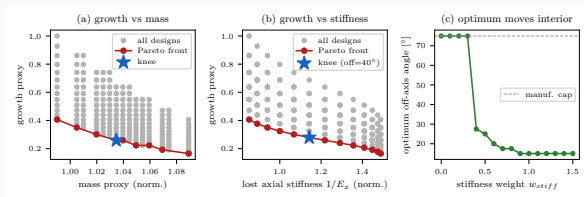
A12 — Stage-4 hierarchical fleet learning



Conjugate-Gibbs partial pooling over a synthetic fleet ($N=20, F=30$):

- recovers φ : $\mathbb{E}[\mu] = -3.10$ (true -3.0),
 $\mathbb{E}[\sigma] = 0.44$ (true 0.45)
- prior SD sharpens $0.39 \rightarrow 0.10$
- fleet-leader: stable clearance in **1.25** vs **2.75** inspections

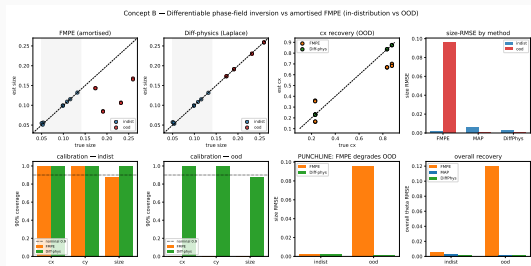
A13 — Stage-5 multi-objective design (Pareto)



Growth vs. mass / axial-stiffness (A_{11}) / bending (D_{11}):

- single growth objective rails off-axis to the manufacturable cap
- competing CLT objectives restore an interior optimum (knee $\sim 40^\circ$)
- optimal angle $75^\circ \rightarrow 15^\circ$ as stiffness weight grows (sensitivity to uncalibrated constants: main deck.)

A14 — Differentiable inversion (Keio bridge)



JAX AT2 forward, FD-verified gradient (3×10^{-7}); Laplace posterior $\Sigma = \sigma^2(J^\top J + \Lambda)^{-1}$, no labelled pairs.

- amortised collapses OOD ($\sim 25\times$, 0 coverage)
- differentiable-physics invariant
- FNO-as-forward: MAP+Laplace 530s $\rightarrow$ 12s ($43\times$)

$\Rightarrow$ the cheap forward TMCMC needs (Keio Phase-3).

A15 — Data provenance and honesty

- **real data**: interstage TSA full-field (Kojima, CPB 2025 — **co-author / consent gated**); fairing long-term + impact guided waves (OGW); SRB-3 AE (4TU CC0); 4TU DCB fatigue.
- **synthetic**: phase-field forward (representative G_c, ℓ, β , fatigue); synthetic fleet; decision-UQ oracle = exact FD physics.
- Reproducibility: ~250 unit tests, one-command end-to-end run, MMS / mesh / SNR / economic-sensitivity studies; figures script-generated.
- **Open gap**: no real life-test of the prognosis→clearance→fleet chain.